

Michael P. Grehan

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PhD in Physics

2023–Present

University of Toronto, Canadian Institute for Theoretical Astrophysics
Supervisors: Christopher Thompson and Bart Ripperda
Expected graduation 2028

MSc in Physics

2022–2023

University of Toronto — GPA: 4.0/4.0

Supervisor: Spyros Alexakis

Thesis: *An Introduction to the AdS/CFT Correspondence and Holographic Entanglement Entropy*

BSc Honours in Physics, Mathematical Specialization

2017–2022

University of Winnipeg — Major GPA: 4.5/4.5 — Cumulative GPA: 4.4813/4.5

Supervisor: Andrew Frey

Thesis: *Complexity of the Klebanov-Strassler Solution*

* Please note that the standard ordering of authors in THEP is alphabetical and that the ordering does not indicate the contributions of an author.

Publications

- 2026 **Michael P. Grehan**, Bart Ripperda, Andrei M. Beloborodov, Christopher Thompson, Elias R. Most, “*Global Magnetohydrodynamic Simulations of Monster Shocks in Neutron Star Magnetospheres*,” [[arXiv:2602.21290](https://arxiv.org/abs/2602.21290) [astro-ph.HE]], published in the Astrophysical Journal, doi: [10.3847/1538-4357/ae633a](https://doi.org/10.3847/1538-4357/ae633a).
- 2026 * A. R. Frey, **M. P. Grehan** and P. Singh, “*Holographic complexity of the Klebanov-Strassler background*,” [[arXiv:2311.18804](https://arxiv.org/abs/2311.18804) [hep-th]], published in Physical Review D, doi: [10.1103/6ft7-phhs](https://doi.org/10.1103/6ft7-phhs).
- 2025 **Michael P. Grehan**, Tanisha Ghosal, James R. Beattie, Bart Ripperda, Oliver Porth, Fabio Bacchini, “*Comparison of magnetic diffusion and reconnection in ideal and resistive relativistic magnetohydrodynamics, ideal magnetodynamics and resistive force-free electrodynamics*,” [[arXiv:2503.20013](https://arxiv.org/abs/2503.20013) [astro-ph.HE]], published in Physical Review D, doi: [10.1103/8xf2-x2nq](https://doi.org/10.1103/8xf2-x2nq).
- 2021 * A. R. Frey, **M. P. Grehan** and M. Srivastava, “*Complexity of Scalar Collapse in Anti-de Sitter Spacetime*,” [[arXiv:2110.09630](https://arxiv.org/abs/2110.09630) [hep-th]], published in the Journal of High Energy Physics, doi: [0.1007/JHEP12\(2021\)135](https://doi.org/10.1007/JHEP12(2021)135).

Preprints

- 2026 B. Ripperda, **M. P. Grehan**, A. Moran, S. Selvi, L. Sironi, A. Philippov, A. Bransgrove, O. Porth, “*Magnetic reconnection with a 0.1 rate: Effective resistivity in general relativistic magnetohydrodynamics*,” [[arXiv:2601.02460](https://arxiv.org/abs/2601.02460) [astro-ph.HE]](Submitted to Physical Review Letters.)

Talks

- 2026 Invited talk at the Princeton astrophysics group meeting: “*Modeling Magnetic Dissipation in Magnetized Magnetospheres*”.
- 2026 Invited talk at the Simons Collaboration on Extreme Electrodynamics of Compact Sources (SCEECS) satellite meeting of the annual meeting: “*Global Magnetohydrodynamic Simulations of Monster Shocks in Neutron Star Magnetospheres*”
- 2025 Invited talk at the Kavli Institute for Particle Astrophysics (KIPAC) and Cosmology tea at Stanford University: “*Modeling Magnetic Dissipation in Magnetar Magnetospheres*”.
- 2025 Contributed talk at Frontiers of Relativistic Plasma Physics in Astrophysics and Laboratory Experiments hosted at the Kavli Institute for Theoretical Physics (KITP): “*Modeling Magnetic Dissipation in Magnetar Magnetospheres*”.
- 2025 Contributed talk at the Perimeter Institute for Theoretical Physics workshop for Magnetic Fields around Compact Objects: “*Magnetic diffusion and reconnection: a comparison between physical models*”.

Posters

- 2025 Ecole de physique des Houches workshop feeling the pull and the pulse of relativistic magnetospheres: “*Magnetic diffusion and reconnection: a comparison between physical models*”.

Conferences, Workshops, and Meetings

- 2026 Attended the Seventh Purdue Workshop on Relativistic Plasma Astrophysics.
- 2026 Attended the Simons Collaboration on Extreme Electrodynamics of Compact Sources Annual Meeting.
- 2025 Attended the Simons Collaboration on Extreme Electrodynamics of Compact Sources Annual Meeting.
- 2025 Contributed a talk and attended the conference Frontiers of Relativistic Plasma Physics in Astrophysics and Laboratory Experiments hosted at the Kavli Institute for Theoretical Physics (KITP).
- 2025 Visited KITP for the program Frontiers of Relativistic Plasma Physics in Astrophysics and Laboratory Experiments, as an affiliate of Christopher Thompson.
- 2025 Contributed a talk and attended the Perimeter Institute for Theoretical Physics workshop for Magnetic Fields around Compact Objects.
- 2025 Contributed a poster and attended the Ecole de physique des Houches workshop feeling the pull and the pulse of relativistic magnetospheres.
- 2024 Attended the Simons Collaboration on Extreme Electrodynamics of Compact Sources (SCEECS) Summer School.

Compute Grants

- 2026 Digital Research Alliance of Canada (Collaborator). Project: *Extreme Electrodynamics and Plasma Physics of Neutron Stars* (PI: Christopher Thompson). Awarded 71,236,320 CPU core-hours, 219,000 GPU hours, and 1.6 PB of storage (approximate total value: \$356,170 CAD).

Awards

- 2024 Canada Graduate Scholarship-Doctoral (CGS D). NSERC offers the CGS D to the highest-scored Postgraduate Scholarships-Doctoral (PGS D) applicants. Value: \$120,000 over 36 months.
- 2022 Chancellor's Gold Medal for the Highest Standing in Science (Honours). The medals are given to the students who secure the highest overall standing determined by the final grades obtained in the best 90 or 120 credit hours completed, as applicable, that satisfy degree requirements.
- 2022 Gold Medal for Achievement in a Major (Honours) Physics. This medal is awarded at Convocation to the student with the highest overall average in the major.
- 2021 Student of Highest Distinction, for achieving a sessional grade point average between 4.00 and 4.50.
- 2021 Academic Proficiency Scholarship (Winnipeg Rh Institute)
- 2021 Sir William Stephenson Scholarship awarded annually to one or two students who demonstrate outstanding academic achievement, superior leadership qualities on or off campus, and the potential to make a valuable contribution to Canada.
- 2021 Brian J. Hyslop Memorial Scholarship in Physics, awarded to a student enrolled at The University of Winnipeg in the honours degree in physics, recipients will have worked under the supervision of a faculty member in the Department of Physics on a research project, further consideration will also be given to students interested in pursuing a master's degree in physics.
- 2021 Randy Kobes Poster Contest Prize.
- 2021 IPP Summer Student Program: Undergraduate Research Experience at CERN & in Canada
- 2021 NSERC Undergraduate Student Research Award to work under the supervision of Dr. Andrew Frey.
- 2020 Student of Highest Distinction, for achieving a sessional grade point average between 4.00 and 4.50.
- 2020 Crawford Campbell Memorial Scholarship - It will be awarded to a continuing student in the General, 4-year, or Honours Mathematics program of scholarly ability and entering Advanced Calculus (now known as Introduction to Mathematical Analysis).
- 2020 William L. Dyker Memorial Scholarship - Awarded annually to a student who has completed 30 credit hours and is continuing at The University of Winnipeg in any undergraduate degree program.
- 2020 Chancellor W. John A. Bulman Scholarship - Awarded to an outstanding student or students in any undergraduate degree program.
- 2020 Academic Proficiency Scholarship (Winnipeg Rh Institute)
- 2020 Randy Kobes Poster Contest Prize - Awarded for placing first in the 15th Annual Randy Kobes Poster Symposium in the Math and Theoretical Physical Science section, presenting my summer 2020 research to a general public audience.

- 2020 Dr. Herbert Shubin Memorial Scholarship in Physics - Awarded annually to the most promising student entering the final year of the 4-year Physics program.
- 2020 Manitoba Scholarship and Bursary Initiative - Scholarship for the Fall/Winter 2019-20 academic year for academic excellence.
- 2020 NSERC Undergraduate Student Research Award to work under the supervision of Dr. Andrew Frey.
- 2019 Student of Highest Distinction, for achieving a sessional grade point average between 4.00 and 4.50.
- 2019 Academic Proficiency Scholarship (Herbert L. Draper Fund).
- 2019 Lawson Scholarship in Mathematics, awarded to a student of outstanding academic promise who is entering or continuing Major work in the Mathematics Major at The University of Winnipeg.
- 2019 Brian J. Hyslop Memorial Scholarship in Physics, awarded to a student enrolled at The University of Winnipeg in the honours degree in physics, recipients will have worked under the supervision of a faculty member in the Department of Physics on a research project, further consideration will also be given to students interested in pursuing a master's degree in physics.
- 2019 Isbister Undergraduate Scholarship, two scholarships are offered to two top students in Arts and two to two top students in Science.
- 2019 Rev. John H. and Mrs. Myrtle (Webster) Shemilt Scholarship, awarded annually to a full-time student in any undergraduate degree program of The University of Winnipeg. Preference will be given to student from Manitoba or northwestern Ontario.
- 2019 NSERC Undergraduate Student Research Award to work under the supervision of Dr. Andrew Frey.
- 2018 Sandra and Harvey Selter Scholarship, awarded to one or more outstanding students in any Faculty who demonstrate a commitment to their community.
- 2018 Henry Doidge Memorial Scholarship in Physics, awarded to a student with standing in first year Physics who is entering the second year of the Physics Major program.
- 2018 Donald S. Grant Memorial Scholarship for Excellence in Mathematics, given to a continuing student in Mathematics and Statistics who has attained a high standing in 6 credit hours of introductory calculus.

Teaching Experience

Cambridge Centre for International Research

- 2026 TA for Black Holes: From General Relativity to Quantum Electrodynamics.
- 2025 TA for The Astrophysics of High Density Objects: Plasma Physics, General Relativity, and Quantum Electrodynamics.
- 2025 TA for Black Holes: From General Relativity to Quantum Electrodynamics.

University of Toronto

- 2026 TA for PHY152: Foundations of Physics II.
- 2025 TA for PHY151: Foundations of Physics I.
- 2025 TA for PHY132: Introduction of Physics II.
- 2024 TA for PHY151: Foundations of Physics I.
- 2024 TA for PHY152: Foundations of Physics II.
- 2023 TA for PHY131: Introduction to Physics I.
- 2023 TA for PHY132: Introduction to Physics II.
- 2022 TA for PHY131: Introduction to Physics I.

University of Winnipeg

2022	TA for PHYS-1101: Foundations of Physics.
2022	TA for PHYS-3301: Quantum Mechanics.
2021	TA for PHYS-2105: Mathematical Physics I.
2021	TA for PHYS-3403: Thermal and Statistical Physics.
2021	TA for PHYS-3203: Advanced Mechanics.
2021	TA for PHYS-1101: Foundations of Physics.
2020	TA for PHYS-3202: Intermediate Mechanics.
2020	TA for PHYS-2302: Modern and Thermal Physics.
2020	TA for PHYS-1101: Foundations of Physics.
2019	TA for PHYS-2201: Electricity and Magnetism.
2019	TA for PHYS-1301: Introduction to Physics.
2019	TA for PHYS-1101: Foundations of Physics.
2018	TA for PHYS-1301: Introduction to Physics.

Outreach and Community Involvement

2026	Visited Anola School to speak with middle school students about post secondary education and astrophysics.
2026 - Present	Member of CITA visitor committee.
2026 - Present	Simons Collaboration on Extreme Electrodynamics of Compact Sources (SCEECS) Affiliated Scientist.
2026	Volunteered to speak about my research to high school students at Physics Day ran by Let's Talk Science (LTS) at the University of Toronto.
2020-2022	Elected Vice President of the University of Winnipeg Physics Student Association (UWPSA).
2017-2022	Member of UWPSA.
2017-2018	Member of the UWPSA Treasury Committee.

Last updated: September 15, 2026